

Full Stack Developer Case Study Submission Report

Project: Mini ERP + CRM Operations Portal

Submission Requirements & Deliverables Check

This document addresses all 8 required submission items specified in the case study guidelines:

Submission Requirement Item	Deliverable Details / Link
1. GitHub Repository Link	https://github.com/bhargavreddy-star/FundsRoom-Mini-ERP-CRM
2. Live Frontend URL	https://fundsroom-mini-erp-crm.vercel.app (Local: http://localhost:5173)
3. Live Backend API URL	https://fundsroom-erp-backend.onrender.com (Local: http://localhost:3001)
4. Test Login Credentials	Admin, Sales, Warehouse, Accounts credentials (see details in Section 3)
5. Postman Collection / API Docs	Included in repo root as <code>postman_collection.json</code> & detailed in Section 5
6. README Setup & Deployment Instructions	Included in repo root as <code>README.md</code> & detailed in Section 6
7. Short Explanation of Architecture	Node/Express + Prisma ORM + SQLite + React Vite CSS (see Section 2)
8. Known Limitations or Incomplete Parts	Documented in Section 7

1. Executive Summary & Project Overview

The FundsRoom Mini ERP + CRM Operations Portal is a full-stack, enterprise-grade web application designed for wholesale and distribution companies. It streamlines business operations across internal teams—Sales, Warehouse, Accounts, and Management—by providing single-source management of customer leads, inventory tracking with stock movement audits, and delivery challan execution with strict stock validation.

2. Short Explanation of Architecture

The application follows a decoupled client-server architecture with clean RESTful API standards:

Backend Architecture

- **Framework & Language:** Node.js & TypeScript with Express.js framework.
- **ORM & Database:** Prisma ORM with SQLite database (dev.db) for type-safe database queries.
- **Security & Auth:** Stateless JSON Web Tokens (JWT) with Bcrypt password hashing.

Frontend Architecture

- **Framework & Bundler:** React 18 + TypeScript built with Vite.
- **Design System:** Custom Vanilla CSS system featuring HSL color tokens, dark-mode glassmorphic cards, micro-animations, and responsive Flexbox/Grid layouts.

3. Test Login Credentials for All Roles

The system implements 4 predefined user roles with tailored workspace navigation and permissions:

Role	Email	Password	Access Level & Scope
Admin	admin@fundsroom.com	password123	Full unrestricted operational & management access
Sales	sales@fundsroom.com	password123	Customer CRM, lead follow-ups & Delivery Challan creation
Warehouse	warehouse@fundsroom.com	password123	Product stock management, Stock Movement logs (IN/OUT)
Accounts	accounts@fundsroom.com	password123	Read-only financial audit oversight for transactions & challans

4. Core Functional Modules

Module 1: Customer CRM

Tracks customer data including Name, Mobile, Email, Business Name, optional GST Number, Customer Type (Retail, Wholesale, Distributor), Address, Status (Lead, Active, Inactive), Follow-up Date, and Notes. Includes real-time search, pagination, and follow-up interaction history.

Module 2: Product & Inventory Management

Tracks Product Name, SKU, Category, Unit Price, Current Stock, Minimum Stock Alert, and Warehouse Location. Includes low-stock threshold alerting (highlighted when stock falls below alert quantity) and Stock Movement logging (IN/OUT movements with user signature, quantity, reason, and timestamp).

Module 3: Sales Delivery Challan Workflow

Enables selecting customers, adding multiple products with quantities, auto-generating unique challan numbers (CH-YYYYMMDD-XXXX), and managing draft/confirmed/cancelled statuses. Validates stock before creation/confirmation (preventing negative inventory) and snapshots product details into delivery items.

5. Postman Collection & API Documentation

A pre-configured Postman Collection JSON file (postman_collection.json) is included in the project repository root. It contains structured requests for Authentication, Customer CRUD, Product Inventory Movements, and Sales Delivery Challan execution.

6. README Setup & Deployment Instructions

Full setup and deployment instructions are provided in the root README.md file:

1. Backend Setup: `cd backend && npm install && npx prisma db push && npx tsx prisma/seed.ts && npm run dev`
2. Frontend Setup: `cd frontend && npm install && npm run dev` (accessible at <http://localhost:5173>)
3. Docker Containerization (Bonus): `docker-compose up --build`
4. Render & Vercel Deployment: Configured via environment variables (PORT, DATABASE_URL, JWT_SECRET, VITE_API_URL).

7. Known Limitations & Trade-offs

- **Database Engine:** Uses SQLite for zero-config local evaluation; production can switch to PostgreSQL by changing provider in schema.prisma.
- **Token Storage:** JWT is stored in localStorage for simple API interceptor handling.